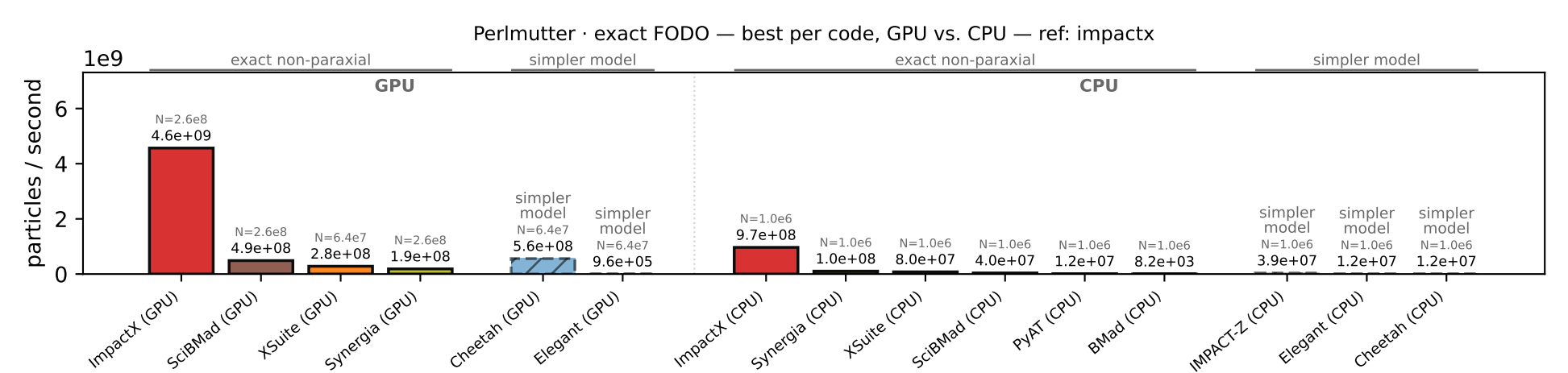

'simpler model' = a lower-fidelity model vs the reference — cheetah: paraxial chromatic-linear quad; elegant: trace-space slopes + achromatic drift; impactz: paraxial chromatic thick quad (transfmapK, K/(1+delta))

each bar = the fastest config for that code on that device (any precision / fast-math); N = the particle count that bar was measured at (top of each device's sweep)

hardware: CPU: AMD EPYC 7763 (64 cores) GPU: NVIDIA A100-SXM4-40GB

versions: impactx 26.08 · scibmad 0.4.1 · xsuite 0.53.4 · synergia v2024.04-209-g795b99f39 · cheetah 0.8.2 · elegant elegant-2026.4.0 · pyat 0.8.0 · bmad 20260904-1 · impactz V2.7.7